

PayShield

Real-Time UPI Payment Verification for Every Vendor

A two-layer fraud shield that tells a vendor — in seconds — whether a UPI payment screenshot is Genuine, Suspicious, or Fake, and why.

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AGENDA

What we'll cover

01

The Problem

Why fake UPI screenshots hurt real vendors

02

Our Solution

PayShield's two-layer verification shield

03

How It Works

Architecture, user journey & tech stack

04

Business & Impact

Model, go-to-market, market sizing

05

Roadmap & Team

What's next, and who's building it

THE PROBLEM

Fake payment screenshots are a growing tax on India's smallest merchants.

India's digital-payments boom has made UPI the default way to pay — but the same convenience that helps honest buyers also gives scammers an easy trick: show a doctored 'Payment Successful' screen, and walk away before the vendor checks their bank account.

1

Editing apps and screen-mockup tools can recreate GPay / PhonePe / Paytm success screens pixel-for-pixel, down to the fonts and icons.

2

Small vendors and street sellers rarely have time to call their bank or check an SMS before handing over goods — the moment of trust is exploited.

3

There is no fast, in-the-moment way for a shopkeeper to independently confirm a screenshot before the buyer leaves.

INDIA'S UPI SCALE

24.5 Bn+

UPI transactions processed in a single month (Aug 2026, NPCI)

55+ Cr

Registered UPI users across India (mid-2026)

6.5 Cr+

Merchants now accepting UPI payments

At this scale, even a small fraud rate translates into large, everyday losses for the merchants who can least afford it.

THE PROBLEM

Anatomy of a fake-screenshot scam

A realistic walkthrough of how the fraud plays out at the counter.

1

Setup

Buyer picks up ₹1,250 worth of goods at a small store and offers to 'pay by UPI.'

2

Forgery

Using a screen-mockup or editing app, the buyer generates a fake 'Payment Successful' screen matching the vendor's UPI app.

3

Deception

The buyer briefly shows the screen, says 'sent, check later,' and leaves quickly — relying on the vendor being busy or trusting.

4

Discovery

The vendor checks their bank balance at closing time and finds no credit ever arrived — by then, the buyer is long gone.

PayShield's goal: break this loop at Step 2 — the moment a fake screenshot is shown — not at Step 4, when it's already too late.

Why this matters at scale

Volume

P2M (person-to-merchant) UPI payments now make up the majority of digital merchant spending in India, and are growing faster than P2P transfers — more merchant-facing transactions means more surface area for screenshot fraud.

Ticket size

The average UPI transaction size has fallen below ₹1,300 as everyday, small-value purchases dominate — exactly the low-friction, low-attention transactions where a fake screenshot is easiest to slip past a busy shopkeeper.

Reach

UPI QR acceptance points now number in the hundreds of millions across India, including tier-2/3 towns and rural markets where vendors have the least access to dispute-resolution support.

THE PROBLEM

Existing approaches fall short

APPROACH	HOW IT WORKS	WHERE IT FALLS SHORT
Manual visual check	Vendor eyeballs the screenshot	Fails against pixel-accurate forgeries; relies entirely on human attention
Wait for bank SMS	Vendor waits for a credit SMS	Slow, easy to rush past; buyer is often gone before it arrives
POS / card machines	Dedicated card terminals	High cost, not viable for street vendors and micro-merchants
Generic screenshot checkers	Third-party ‘fake or real’ apps	Not UPI-specific; no live transaction cross-check; no vendor workflow

SECTION 02

Our Solution

Introducing PayShield — a two-layer verification shield built for the vendor's counter.

PayShield: verification at two levels of defense

01

Layer 1 — Real-Time Verification

The primary path. The vendor shares a PayShield UPI collect-link or QR code. When the buyer pays, confirmation comes directly from the UPI switch (NPCI rails) — not from a screenshot — so it physically cannot be faked.

02

Layer 2 — Screenshot Forensics

The fallback path. If a buyer still shows a screenshot (e.g. paid from their own separate UPI app), PayShield analyses the image with OCR, template matching, metadata checks and an on-device ML model.

Real-time verification: truth from the source

Instead of trusting an image, PayShield asks the payment network itself.

1

Vendor generates request

One tap in the PayShield app creates a UPI collect-request / dynamic QR for the exact bill amount.

2

Buyer approves in their app

The buyer scans or accepts the request in GPay, PhonePe, Paytm or any UPI app — approving from their own bank.

3

Switch confirms, not a screenshot

Settlement confirmation flows back through the UPI switch directly to PayShield's backend — the source of truth.

4

Instant, tamper-proof verdict

Vendor sees a live 'Payment Received ✓' the moment funds move — nothing the buyer shows on their phone can change it.

WHY THIS CAN'T BE FAKED

- Confirmation originates from the UPI switch / bank rails — an external, trusted system the buyer cannot edit.
- PayShield never relies on anything rendered on the buyer's screen for this path.
- Works with any UPI app the buyer already has — no new app required on the buyer's side.
- Falls back gracefully to Layer 2 if the buyer insists on paying via a personal transfer and showing proof instead.

Screenshot forensics: catching the fake at a glance

1

OCR Extraction

Google ML Kit / Tesseract pulls amount, transaction ID, timestamp, and payer/payee UPI ID directly from the image.

2

Template Matching

Compares fonts, icon placement, spacing and layout against a reference library of genuine GPay / PhonePe / Paytm / BHIM screens.

3

Metadata Forensics

Checks image EXIF data, compression artifacts, and edit-history signatures left behind by editing or screen-mockup apps.

4

Transaction Cross-Check

Where possible, matches the extracted transaction ID against the vendor's bank-SMS credit or a UPI API response.

5

On-Device ML Classifier

A lightweight TensorFlow Lite model combines all four signals into one confidence score — fast enough to run without internet.

OUTPUT VERDICT

✓ Genuine

All signals consistent with a real, confirmed transaction.

! Suspicious

Minor mismatch found — vendor advised to verify further.

✗ Fake

Clear signs of editing or a fabricated screen.

*Every verdict comes with a plain-language reason
— never just a red or green label.*

HOW IT WORKS

System architecture



Every component is designed to work in isolation when offline — only the real-time verification path requires connectivity.

The vendor's journey with PayShield

1

Onboard

Vendor downloads the app, verifies their shop via mobile number and links their existing UPI ID — under 2 minutes, no hardware needed.

2

Set up counter

A PayShield QR sticker or in-app dynamic QR is placed at the counter for real-time verification requests.

3

Accept payment

Buyer pays via the QR (Layer 1) or shows a screenshot for instant forensic scanning (Layer 2).

4

Get verdict

Vendor sees Genuine / Suspicious / Fake with a reason, and decides whether to release goods.

5

Review & learn

Daily dashboard shows scan history, flags repeat offenders, and totals fraud amount prevented.

HOW IT WORKS

Why PayShield wins

Built for the vendor who can't afford to be scammed twice.

Works Offline

On-device OCR + template check runs without internet — built for shops with patchy connectivity.

Multi-App Coverage

Detects edited screenshots from GPay, PhonePe, Paytm, and BHIM alike.

Bank-SMS Cross-Check

Matches the screenshot's claim against the vendor's actual bank credit SMS as a second, independent signal.

Vendor Dashboard

Logs every scan, flags repeat offenders, and shows a daily fraud-prevented summary.

Regional Language UI

Tamil / Hindi / English interface so every shopkeeper can use it comfortably.

Privacy-First

Screenshots are processed and discarded — only the verdict and reason are stored.

HOW IT WORKS

Tech stack — lightweight, on-device, built to scale

FRONTEND

Flutter

Single codebase for Android & iOS vendor app; works well on low-end devices common with small merchants.

OCR / VISION

Google ML Kit + Tesseract

On-device text and layout extraction from screenshots, no server round-trip required.

ML MODEL

TensorFlow Lite

Lightweight classifier for the Genuine/Suspicious/Fake verdict, runs directly on the vendor's phone.

BACKEND

Python (FastAPI) + Node.js

Handles collect-requests, vendor accounts, and orchestration between services.

DATABASE

Firebase / PostgreSQL

Firebase for real-time sync on mobile; PostgreSQL for structured transaction and audit records.

PAYMENTS

UPI Deep Linking + NPCI Sandbox APIs

Integrates with the UPI ecosystem for collect-requests and settlement confirmation.

HOW IT WORKS

Security, privacy & compliance by design



Data Minimization

Screenshots are processed in-memory and discarded immediately after a verdict is produced — only the verdict, reason, and transaction metadata are stored.



Encryption

All data in transit uses TLS; sensitive vendor and transaction records are encrypted at rest in the database.



DPDP Act Alignment

Data handling is designed around India's Digital Personal Data Protection Act principles — purpose limitation, consent, and minimal retention.



No Card / PIN Storage

PayShield never asks for or stores UPI PINs, card numbers, or OTPs — all payment authorization happens inside the buyer's own UPI app.



Auditability

Every verdict is logged with a timestamp and reason, giving vendors (and, if needed, banks) a clear audit trail for disputes.

Business model: free to start, sustainable to scale

FREE

Solo vendors

- Screenshot forensics (Layer 2)
- Basic dashboard
- Up to 50 scans / day

PRO

Growing shops

- Everything in Free
- Real-time QR verification (Layer 1)
- Bank-SMS cross-check
- Unlimited scans

ENTERPRISE

Retail chains & banks

- API access for integration
- Multi-branch dashboard
- Fraud analytics & alerts
- Dedicated support

Revenue also flows from optional bank & fintech partnerships — white-labelled fraud-verification APIs for wallets, POS providers, and payment aggregators.

Go-to-market strategy

1

Phase 1 — Pilot

Launch in 2–3 local markets (street vendor clusters, college-town shops) with hands-on onboarding and free-tier access.



2

Phase 2 — Community Growth

Vendor-to-vendor referrals, local trader-association tie-ups, and market-association demos to drive organic adoption.



3

Phase 3 — Partnerships

Integrate with UPI apps, POS providers, and local banks as a plug-in fraud-verification layer for their existing merchant base.

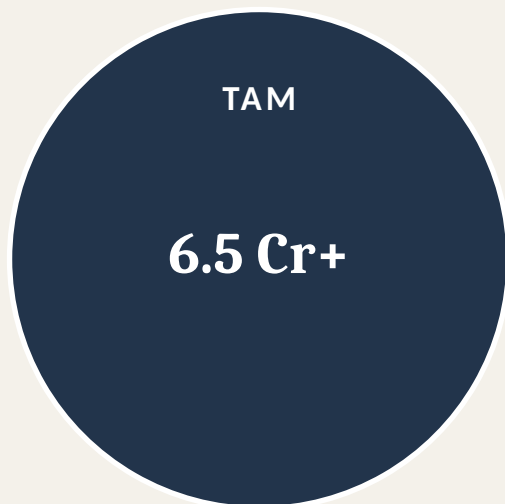


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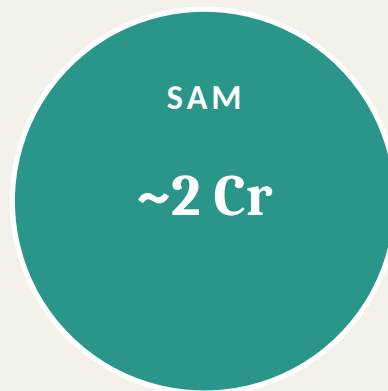
Phase 4 — Scale

Expand regional-language support and roll out Enterprise API access for retail chains and payment aggregators.

Market opportunity



All UPI-accepting merchants in India



Small & micro merchants doing frequent in-person
UPI sales (street vendors, kirana stores, local
services)



Realistic 3-year target: vendors reachable via city
pilots, trader associations & partner banks

What's next

Now — Hackathon

Working prototype: screenshot forensics (Layer 2) with OCR + template matching + verdict UI.



Q1

Pilot Layer 1 real-time verification with a small group of local vendors; refine ML classifier accuracy.



Q2

Add bank-SMS cross-check, vendor dashboard analytics, and Tamil / Hindi language support.



Q3-Q4

Partner integrations (UPI apps, POS providers), Enterprise API launch, expand to new cities.

OUR TEAM

The people behind PayShield



M1

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[Team Lead / Backend]



M2

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Thank You

PayShield — Spark. Build. Impact.

Questions & Feedback Welcome